Rebuilding Weekly Dengue Curves from Annual Totals: Results, Ablation Study and Discussion

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Abstract

We rebuild the weekly dengue curve of a municipality from its known annual total and ask which information improves the rebuilt shape. On 300 Brazilian municipalities (2010–2023, 3,194 held-out municipality-years in the main design), local weather lowers the shape error by only 1.3 % relative to a seasonal Dirichlet-multinomial baseline (B4s), and the base paper's scaled INLA by 5.3 %. Adding same-year Google Trends search interest lowers it by 12.7 % (95 % year-bootstrap interval [-17.9, -8.4] %), better in 12 of 12 test years, and the gain replicates in a second validation design (12.8 %, 14 of 14 years) and at every training length from two to thirteen years (12.0–22.7 % with $k = 2$–9 training years). Ablations show that the gain comes from the timing of the right year (it disappears with another year's searches), that weather adds nothing once search data are included, and that the gain is smallest in the North and in states with few searches. All numbers, tables and figures in this document are generated by one script from the committed run outputs (Section 6).

1 Problem and research gaps

Many dengue records are annual, while planning, forecasting and climate–health studies need weekly curves. The base paper (Lim et al., 2026) splits annual totals into sub-annual shares with a Dirichlet-multinomial (DM) model and selects scaled INLA as its final method; in both, every rebuilt year of a place receives essentially the same seasonal shape, so epidemics that start early or late are missed. Related work either validates against synthetic or single-year data (Morello et al., 2026; Al Mobin et al., 2023, 2025), adds only climate (Rozo Posada et al., 2026), or uses search data for nowcasting rather than for rebuilding past years (Yang et al., 2017; Xiao et al., 2025). This document reports the experiments that address these gaps: (G1) a year-specific shape instead of one fixed shape; (G2) validation against true weekly counts over many years; (G3) a head-to-head comparison of climate with a same-year search signal; and (G4) component-level evidence of where the gain comes from.

2 Experimental setup

Data. Weekly notified dengue cases (InfoDengue) for 300 municipalities in all 27 states, 2010–2023; NASA POWER weather (CHIRPS rain as a check); the Oceanic Niño Index (ONI); weekly Google Trends interest for "dengue" for Brazil and each state (five overlapping four-year windows, 140 files, kept as a snapshot). The year 2024 is locked and never loaded.

Task and metric. The annual total N of a held-out municipality-year is known; a model predicts the share of N in each epidemiological week, and the counts are allocated to sum exactly to N. The primary metric is the Jensen–Shannon divergence (JSD) between the true and rebuilt weekly shares (0 = identical; multinomial noise floor ≈ 0.015). Secondary metrics: peak-week error and onset-week error (week in which 10 % of the cases are reached), in weeks. "Change %" is the relative change in mean JSD; negative means less error.

Table 1: Mean shape error (JSD, lower is better) of every model on design A (expanding window, 12 test years, 3,194 municipality-years) and design C (leave one year out, 14 folds). Negative "vs B4s" = less error than the B4s baseline. Scaled INLA was run on design A only.

Model	Description	JSD (A)	95% CI (towns)	vs B4s %	Peak err (wk)	Onset err (wk)	JSD (C)	vs B4s % (C)
B0	Uniform weeks	0.1861	[0.1797, 0.1926]	+56.4	11.74	–	0.1888	+70.0
B1	Seasonal profile	0.1282	[0.1230, 0.1337]	+7.7	6.27	3.22	0.1197	+7.8
B4	Base paper's DM	0.1256	[0.1204, 0.1310]	+5.6	6.22	–	0.1191	+7.2
B4s	DM + annual size (baseline)	0.1190	[0.1137, 0.1247]	–	6.41	2.99	0.1110	–
M1-noONI	B4s + local weather	0.1175	[0.1121, 0.1232]	-1.3	6.25	2.91	0.1077	-3.0
M1	B4s + weather + ONI	0.1191	[0.1138, 0.1247]	+0.1	6.26	2.90	0.1090	-1.8
M1-short	B4s + weather, lags 0–3 wk	0.1166	[0.1114, 0.1223]	-2.0	6.30	–	0.1089	-1.9
M2	Boosted softmax (weather)	0.1267	[0.1216, 0.1321]	+6.5	6.21	–	0.1182	+6.4
INLA-scaled	Scaled INLA (base paper's final)	0.1126	[0.1084, 0.1171]	-5.3	5.74	–	–	–
INLA-scaled-clim15	Scaled INLA + local weather	0.1110	[0.1068, 0.1154]	-6.7	5.68	–	–	–
INLA-scaled-clim20	Scaled INLA + weather + ONI	0.1131	[0.1088, 0.1175]	-4.9	5.70	–	–	–
T1	B4s, seasonal shape shifted	0.1096	[0.1039, 0.1155]	-7.9	5.70	2.55	0.1041	-6.2
S2	B4s + Google Trends	0.1039	[0.0985, 0.1096]	-12.7	5.77	2.57	0.0968	-12.8
S2C-noONI	S2 + local weather	0.1056	[0.1003, 0.1113]	-11.2	5.80	2.58	0.0968	-12.8
S2C	S2 + weather + ONI	0.1073	[0.1019, 0.1129]	-9.9	5.84	2.60	0.0979	-11.8
S1	B4s + state curve (upper bound)	0.0971	[0.0920, 0.1026]	-18.4	5.43	2.44	0.0871	-21.6
S3	S1 + Google Trends (upper bound)	0.0963	[0.0912, 0.1018]	-19.1	5.41	2.41	0.0868	-21.8
S3C	S3 + weather + ONI (upper bound)	0.0991	[0.0940, 0.1046]	-16.7	5.45	2.44	0.0873	-21.4

Upper-bound rows need the same year's weekly counts of other towns in the state, which a truly missing year does not have; they are reported as a ceiling, not as a usable method.

Table 2: Paired bootstrap tests (2,000 replicates) of the main contrasts in both designs. % = relative change in mean JSD of the first model vs the second; "years" = test years in which the first model had the lower mean JSD.

Contrast	A: %	A: CI towns	A: CI years	A: years	C: %	C: CI towns	C: CI years	C: years
S2 vs B4s	-12.7	[-13.9, -11.6]	[-17.9, -8.4]	12/12	-12.8	[-14.2, -11.4]	[-18.8, -7.5]	14/14
S2 vs M1-noONI	-11.6	[-12.6, -10.6]	[-15.3, -8.5]	12/12	-10.1	[-11.4, -8.9]	[-15.4, -5.0]	12/14
S2C-noONI vs S2	+1.7	[+1.2, +2.2]	[-0.8, +4.3]	4/12	+0.1	[-0.3, +0.4]	[-1.0, +1.2]	6/14
T1 vs B4s	-7.9	[-9.4, -6.4]	[-13.3, -3.0]	11/12	-6.2	[-7.5, -4.9]	[-10.8, -2.0]	12/14
S1 vs B4s	-18.4	[-19.8, -17.0]	[-24.2, -13.5]	12/12	-21.6	[-23.1, -20.0]	[-28.3, -14.9]	14/14
S3 vs S1	-0.8	[-1.1, -0.6]	[-1.4, -0.5]	11/12	-0.3	[-0.6, +0.0]	[-1.2, +0.8]	11/14

Validation designs. Design A (primary): expanding window, train on 2010 to $T-1$, test $T = 2012, \ldots, 2023$ (12 folds). Design C: leave one year out (14 folds, 3,723 rows). Design B: fixed test years (2019, 2021, 2023) with k preceding training years (learning curve). All models share folds, municipalities and evaluation rows.

Models. B0 (uniform), B1 (seasonal profile), B4 (base paper's DM), B4s (B4 + annual-size terms; the comparator), M1/M1-noONI/M1-short (B4s + weather anomalies at lags 0–20 weeks, with/without ONI, or lags 0–3 only), M2 (boosted softmax with weather), scaled INLA (base paper's final model, ± weather). Timing models add same-year signals to B4s: S2 (Google Trends anomalies, lags 0–3), S2C (S2 + weather), T1 (seasonal shape shifted by the estimated timing shift), and, as an upper bound only, S1 (curve of the other towns of the same state in the same year, target left out), S3 (S1 + S2) and S3C (S3 + weather). S1/S3 need same-year weekly counts of neighbours, which a truly missing year does not have.

Statistics. Paired bootstraps (2,000 replicates, seed 1) resampling municipalities or test years; a difference is called clear only if both intervals exclude zero.

3 Results

Table 1 and Figure 1 compare all models. Weather moves the error by about 1 %; scaled INLA reaches 0.1126 (5.3 % below B4s); the search model S2 reaches 0.1039, 12.7 % below B4s and 7.8 % below scaled INLA. Only the state-curve upper bound is lower (0.0971).

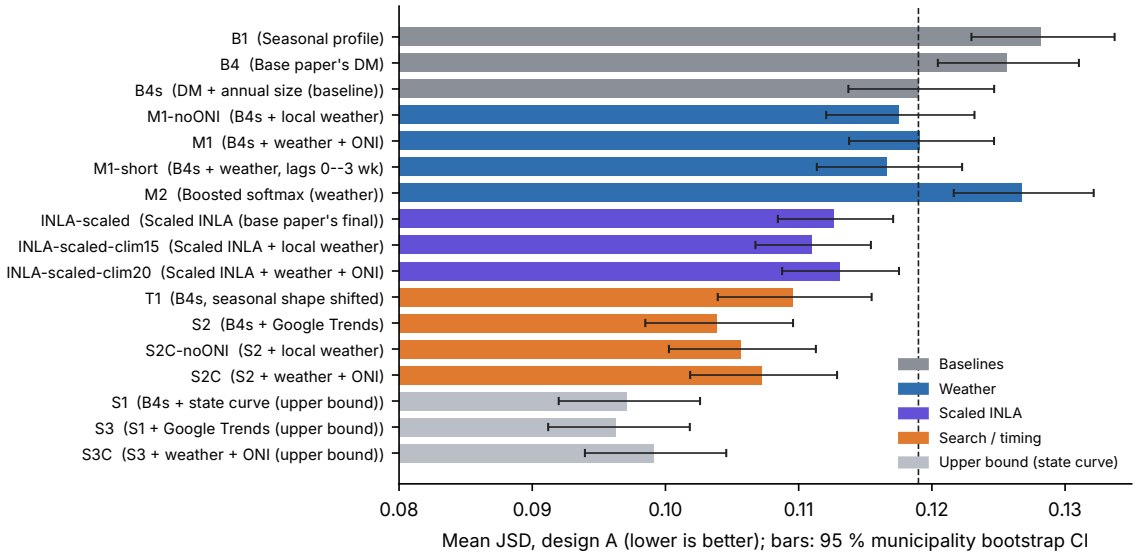


Figure 1: Mean JSD of every model in design A with 95 % municipality bootstrap intervals; dashed line: B4s.

Is the search gain significant? Table 2 gives the paired tests. S2 is better than B4s with both intervals well below zero in both designs (design A: 12.7 % less error, municipality interval [-13.9, -11.6] %, year interval [-17.9, -8.4] %; better in 12 of 12 years; design C: 14 of 14). It is also clearly better than the weather model M1-noONI. At the level of single municipality-years, S2 has the lower error in 72 % of the 3,194 cases (Figure 3, left).

Timing. Figure 2 shows that the search model is below the baseline and the weather model in every test year. In practical units (Figure 4), the mean peak-week error falls from 6.41 (B4s) to 5.77 weeks (S2) and the onset error from 2.99 to 2.57 weeks; the share of municipality-years whose peak is placed within two weeks rises from 29 % to 38 %.

Training length. The learning curve of the climate study (design B, Figure 6) shows that all models improve once each municipality has at least three training years, and that the weather model needs about five years before it stops hurting. With two years every model falls back on regional profiles, so $k = 2$ is out of line.

We ran the same learning curve with the search models, from one training year (train 2010, test 2011) to thirteen (Table 3, Figure 5). Google Trends lowers the error at every training length from two years on: by 12.0 % with only two training years and by 12.0–22.7 % for $k = 2$–9, with all municipality intervals below zero. Weather, in contrast, is harmful with two training years (+11.3 %) and helps by about 3 % from $k \geq 4$; adding weather to Trends is harmful with few years (+9.8 % at $k = 2$) and neutral from $k \geq 6$. With a single training year the search model reduces exactly to B4s: its features are anomalies relative to the training years, which are all zero when there is only one training year, so nothing can be learned. The search signal therefore needs at least two years of history, but not more, which matters for countries with short weekly records.

Integrity checks. Every fold passed the leakage audit and the feature audit, which recomputes the search and state-curve climatologies after replacing all non-training years with random numbers (Table 4).

4 Ablation study

Table 5 and Figure 7 isolate one component per row (design A, paired).

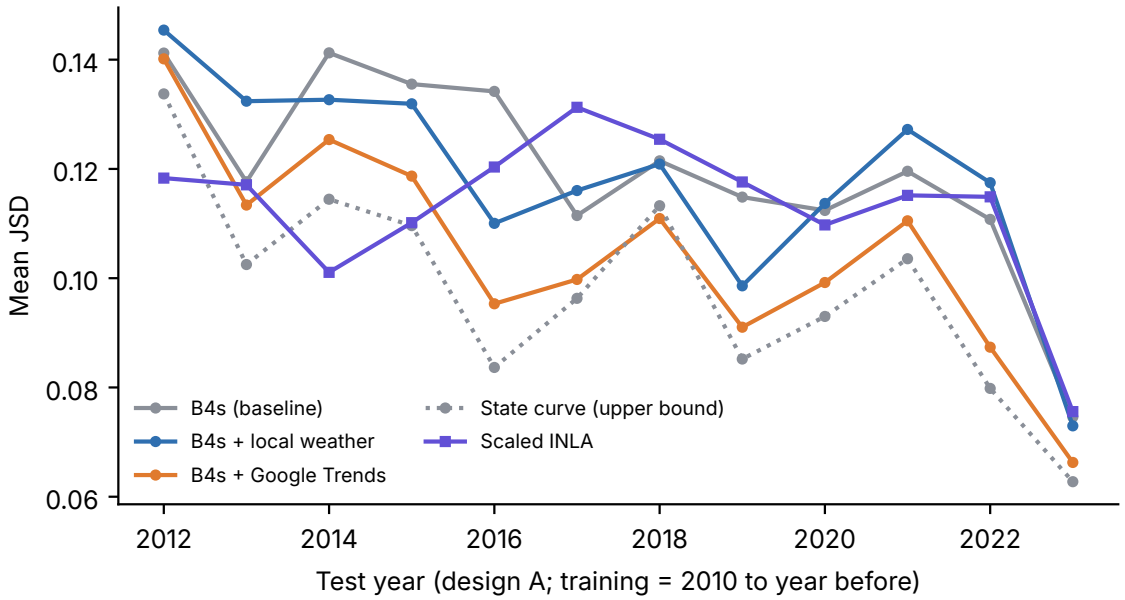


Figure 2: Mean JSD by test year in design A. The training window grows with the test year.

Table 3: Learning curve with Google Trends. k = number of training years immediately before the test year. $k = 1$: train 2010, test 2011. $k = 2$–9: all three fixed test years (2019, 2021, 2023); $k \geq 10$: fewer test years remain, so these rows are not comparable with the others.

k	Test years	B4s	S2 (Trends)	Trends vs B4s %	95% CI towns	Weather vs B4s %	Weather on Trends %
1	2011	0.1014	0.1014	+0.0	[+0.0, +0.0]	+7.2	+7.2
2	2019 2021 2023	0.1056	0.0929	-12.0	[-14.1, -10.0]	+11.3	+9.8
3	2019 2021 2023	0.1424	0.1101	-22.7	[-25.0, -20.3]	-1.5	+3.0
4	2019 2021 2023	0.1184	0.0986	-16.8	[-18.9, -14.7]	-2.9	+1.0
5	2019 2021 2023	0.1112	0.0932	-16.2	[-18.4, -13.9]	-2.8	+1.1
6	2019 2021 2023	0.1095	0.0915	-16.4	[-18.6, -14.3]	-3.2	+0.3
7	2019 2021 2023	0.1071	0.0896	-16.3	[-18.5, -14.2]	-3.9	+0.3
8	2019 2021 2023	0.1045	0.0881	-15.8	[-17.8, -13.7]	-3.4	+0.0
9	2019 2021 2023	0.1027	0.0881	-14.1	[-16.0, -12.3]	-3.4	-0.8
10	2021 2023	0.0959	0.0863	-10.0	[-12.2, -7.7]	+3.4	+1.3
11	2021 2023	0.0954	0.0863	-9.5	[-11.5, -7.4]	+3.4	+2.4
12	2023	0.0743	0.0658	-11.5	[-14.4, -8.5]	-1.1	-0.4
13	2023	0.0746	0.0663	-11.2	[-13.9, -8.4]	-2.2	-1.3

Year-bootstrap intervals are not given: at most three test years per row.

- **Annual size** (B4s vs B4) is the most useful element of the baseline; **weather** gives a small municipality-level gain that is not clear across years; **ONI** and **long lags** make it worse; a **flexible learner** (M2) is worse than the DM.
- **Same-year timing** is the decisive component: even a pure shift of the seasonal shape (T1) is clearly better than B4s, and the search block (S2) roughly doubles that gain.
- **Weather on top of search** does not help (+1.7 %, better in only 4 of 12 years); adding ONI on top is clearly worse. Search on top of the state curve adds less than 1 %.
- **Features vs shift:** entering the state curve as anomaly features (S1) is clearly better than shifting the seasonal shape (T1), so the signal carries more than a single timing offset.
- **Fewer reporting towns:** a state curve from a random 50 % or 25 % of towns keeps most of the gain; from non-study towns only it still beats B4s by a wide margin.
- **Wrong-year controls:** refitting with another year's searches or state curve removes the gain (S2-wy: 0.1185, the level of B4s; Figure 8); feeding another year's signals at test time only

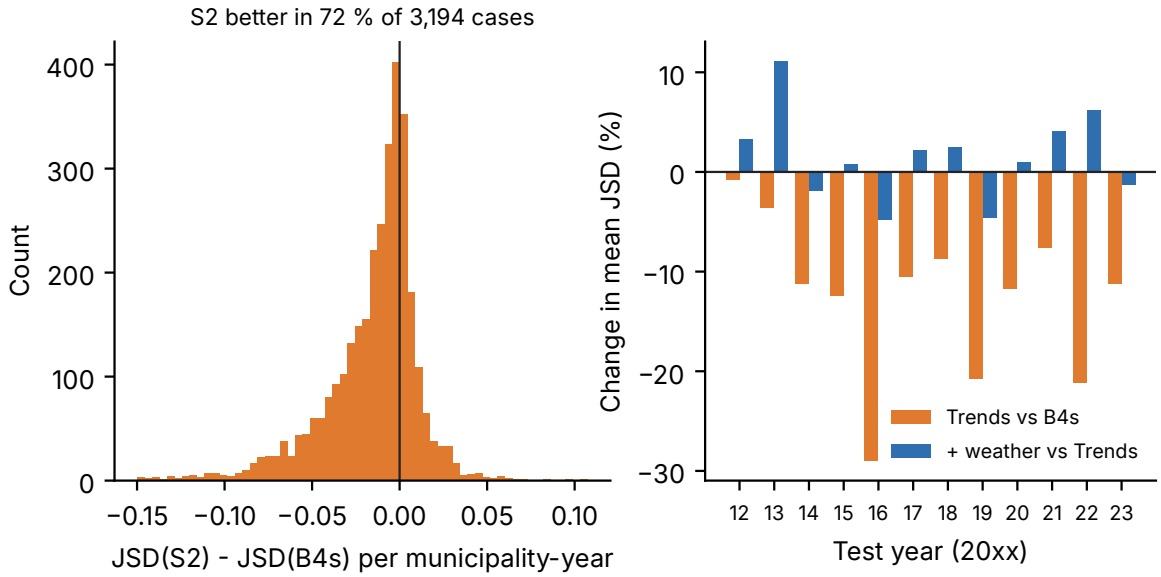


Figure 3: Left: distribution of the paired difference JSD(S2) − JSD(B4s) over municipality-years (values clipped at ±0.15). Right: change in mean JSD per test year for Trends vs B4s and for adding weather to Trends.

Table 4: Automatic checks of the timing-model runs, from the run check files.

Design	Folds	Leakage audit	Feature audit (train-only)	Exact sums
A	12	pass	pass	pass
C	14	pass	pass	pass

makes the model worse than the seasonal baseline (0.1320). The gain therefore comes from the right year's timing, not from model capacity.

5 Discussion

Comparison with baselines. The base paper's scaled INLA improves on the DM baseline by 5.3 %, and weather adds about 1 % to either model. A search signal of the same year improves the DM by 12.7 %, more than twice the INLA gain, with a much tighter interval across years. Weather and climate indices describe the *average* season, which the seasonal terms already capture; searches rise and fall with the year's actual epidemic and therefore carry its timing (gap G3).

Failure cases. The gain is uneven (Table 6, Figure 9). It is largest in the South and Southeast and smallest in the North (about 2 %). By state, Trends is worse than B4s in 4 of 27 states (PI, AP, AC, RR), most of them states with very few searches before 2013, where the search series is mostly zeros with isolated spikes and the national series is used as a fallback for empty windows. The gain also grows with epidemic size (Table 7): in small years the search signal is weak. Figure 10 shows typical cases, including a failure in which a late-December search spike (the start of the next season) pulls the rebuilt curve to the end of the year.

Limitations. (i) The search hypotheses were formulated after the pre-registered climate results, so they are exploratory. (ii) Google Trends starts in 2004 and is sparse in small states, which bounds where the method applies. (iii) Scaled INLA with search data has been implemented and tested

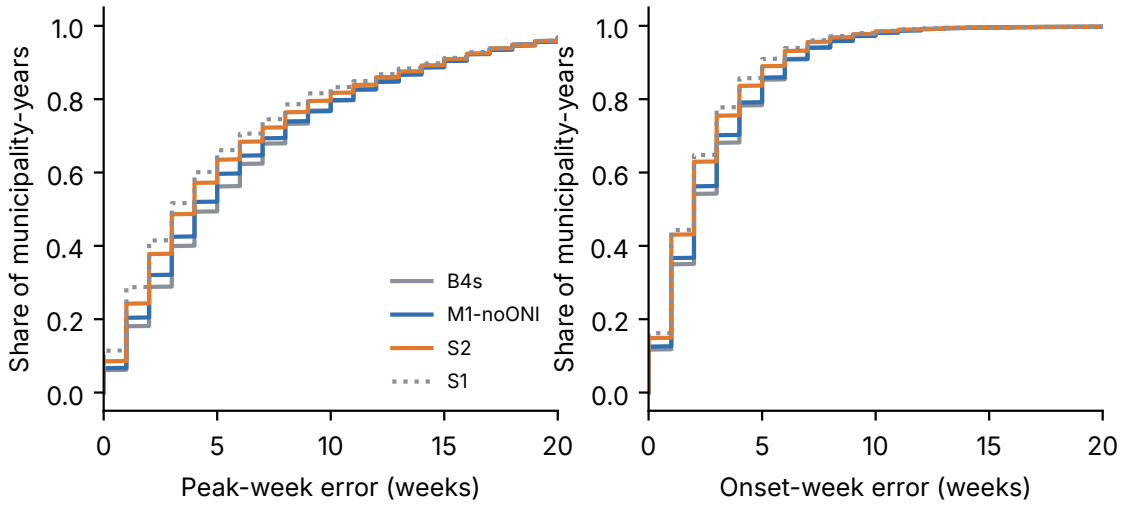


Figure 4: Cumulative distribution of peak-week (left) and onset-week (right) errors, design A. Curves further up and left are better.

on one fold but not yet run on all folds, so the best overall model is not yet known; INLA is also available for design A only. (iv) Peak-week error remains about 5.8 weeks on average: timing is improved, not solved. (v) The learning curve has at most three test years per training length, so it is descriptive (no year-level intervals); with one or two training years the models fall back on regional profiles, and with one year the search effect cannot be learned at all. (vi) The state-curve models are upper bounds, valid only where neighbours report weekly in the same year.

Relation to the research gaps. G1: same-year signals give each year its own shape, and the wrong-year controls show that this is what produces the gain. G2: all results are scored against true weekly counts over 12 and 14 held-out years. G3: in the same folds, search data beat climate by an order of magnitude and make climate redundant. G4: the ablation separates size, climate, timing shift, search and state-curve components, and the learning curve shows how much history each component needs.

6 Reproducibility

Code: `https://github.com/aminah9945/dengue-climate-reconstruction`, branch `results-report`. All tables, figures and the numbers quoted in the text (`report/numbers.tex`) are written by one script that reads only the committed result files; nothing is typed by hand.

```
pip install -e ".[all]"
# 1. data (InfoDengue, NASA POWER, ONI; about 30-45 min)
python scripts/run_pipeline.py --step data
# 2. official climate study (designs A, B, C)
python scripts/run_pipeline.py --step official --design A --climate power --subset 300
python scripts/run_pipeline.py --step official --design B --climate power --subset 300
python scripts/run_pipeline.py --step official --design C --climate power --subset 300
# 3. scaled INLA (R + R-INLA 26.8.7), design A
python -m dengue_recon.extensions.inla run --subset 300 --variants base,clim20,clim15
# 4. timing models (Google Trends snapshot in data_snapshots/google_trends)
export TIMING_TRENDS_DIR=$PWD/data_snapshots/google_trends
python -m dengue_recon.extensions.timing.run --design A --subset 300
python -m dengue_recon.extensions.timing.run --design C --subset 300
python -m dengue_recon.extensions.timing.run --design A --models B4s,M1-noONI,S2,S2C,S2C-noONI \
       --out results/extensions/timing/runs_s2c
python -m dengue_recon.extensions.timing.run --design C --models B4s,M1-noONI,S2,S2C,S2C-noONI \
       --out results/extensions/timing/runs_s2c
# 4b. learning curve with search data (design B, k = 2..13; and k = 1: train 2010, test 2011)
```

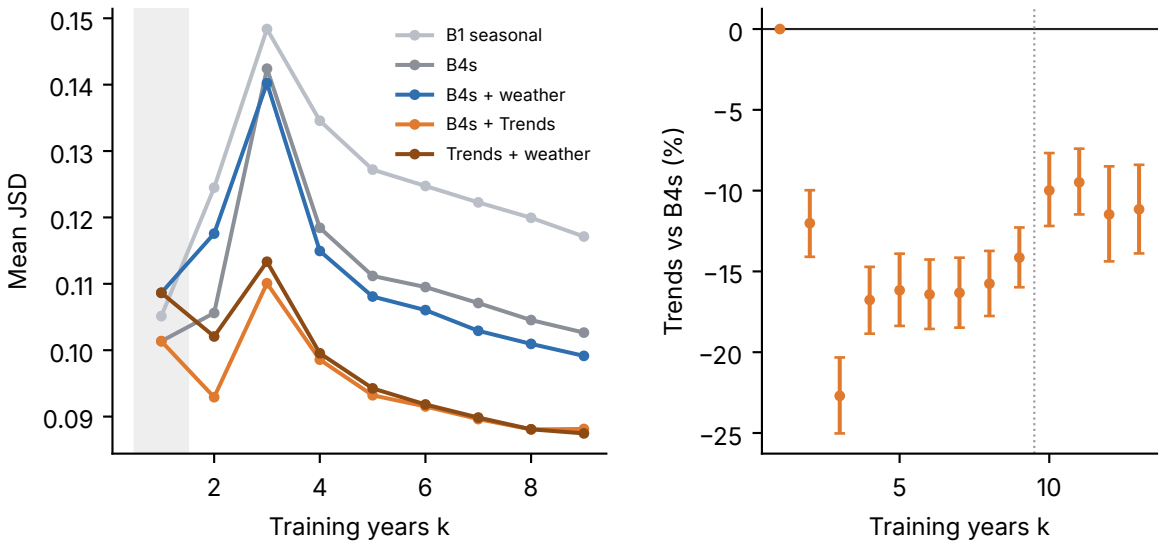


Figure 5: Learning curve with Google Trends. Left: mean JSD by number of training years k (grey band: $k = 1$, test year 2011). Right: change in mean JSD of B4s + Trends vs B4s with 95 % municipality bootstrap intervals; right of the dotted line fewer test years remain.

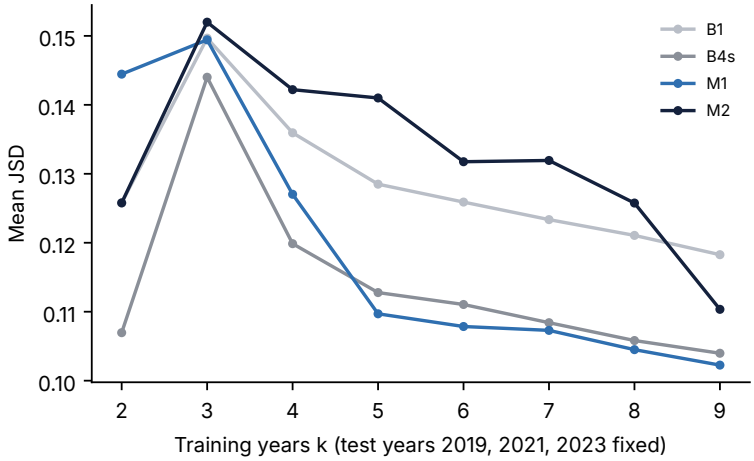


Figure 6: Learning curve (design B): mean JSD against the number of training years, averaged over the three fixed test years. From the official climate run.

```
python -m dengue_recon.extensions.timing.run --design B --out results/extensions/timing/runs_lc
python -m dengue_recon.extensions.timing.run --design A --years 2011 \
       --models B1,B4s,M1-noONI,S2,S2C-noONI --out results/extensions/timing/runs_lc
# 5. every table, figure and number of this report
python scripts/make_results_report.py
```

Colab notebooks `notebooks/06_official_experiments.ipynb`, `07_timing_model.ipynb` and `08_google_trends_download.ipynb` run the same steps. Run logs, per-fold checkpoints and check files are under `results/`.

Table 5: Ablation study, design A: each row adds, removes or alters one component and reports the paired change in mean JSD (negative = the component lowers the error). "Clearly" requires both the municipality and the year bootstrap intervals to exclude zero.

Component changed	Models	JSD with	JSD without	Change %	95% CI towns	95% CI years	Years better	Verdict
Annual size terms	B4s vs B4	0.1190	0.1256	-5.3	[-6.6, -4.0]	[-10.1, +0.9]	10/12	not clear
Local weather	M1-noONI vs B4s	0.1175	0.1190	-1.3	[-2.0, -0.6]	[-6.6, +3.7]	6/12	not clear
ONI on top of local weather	M1 vs M1-noONI	0.1191	0.1175	+1.4	[+0.8, +1.9]	[-0.8, +4.6]	5/12	not clear
Long lags (0–20 vs 0–3 wk)	M1 vs M1-short	0.1191	0.1166	+2.1	[+1.6, +2.6]	[-1.5, +5.8]	3/12	not clear
Boosted learner instead of DM	M2 vs B4s	0.1267	0.1190	+6.5	[+5.2, +7.8]	[-0.5, +12.5]	2/12	not clear
Scaled INLA instead of B4s	INLA-scaled vs B4s	0.1126	0.1190	-5.3	–	[-11.7, +1.4]	7/12	not clear
Shift-only timing (T1)	T1 vs B4s	0.1096	0.1190	-7.9	[-9.4, -6.4]	[-13.3, -3.0]	11/12	clearly better
Google Trends block (S2)	S2 vs B4s	0.1039	0.1190	-12.7	[-13.9, -11.6]	[-17.9, -8.4]	12/12	clearly better
Local weather on top of Trends	S2C-noONI vs S2	0.1056	0.1039	+1.7	[+1.2, +2.2]	[-0.8, +4.3]	4/12	not clear
ONI on top of Trends + weather	S2C vs S2C-noONI	0.1073	0.1056	+1.6	[+1.1, +2.0]	[+0.2, +4.0]	2/12	clearly worse
State-curve block (S1)	S1 vs B4s	0.0971	0.1190	-18.4	[-19.8, -17.0]	[-24.2, -13.5]	12/12	clearly better
Trends on top of state curve	S3 vs S1	0.0963	0.0971	-0.8	[-1.1, -0.6]	[-1.4, -0.5]	11/12	clearly better
Weather + ONI on top of S3	S3C vs S3	0.0991	0.0963	+2.9	[+2.4, +3.5]	[+1.4, +4.4]	3/12	clearly worse
State curve as features vs shift	S1 vs T1	0.0971	0.1096	-11.4	[-12.6, -10.2]	[-14.4, -8.2]	12/12	clearly better
State curve from 50 % of towns	S1-p50 vs S1	0.0984	0.0971	+1.4	[+0.9, +2.0]	[+0.8, +2.0]	1/12	clearly worse
State curve from 25 % of towns	S1-p25 vs S1	0.0996	0.0971	+2.6	[+1.8, +3.3]	[+1.5, +3.9]	1/12	clearly worse
State curve, non-study towns only	S1-ns vs S1	0.1026	0.0971	+5.7	[+4.3, +7.1]	[+3.9, +7.8]	0/12	clearly worse
Wrong-year Trends (refitted)	S2-wy vs S2	0.1185	0.1039	+14.1	[+12.8, +15.7]	[+8.7, +21.2]	1/12	clearly worse
Wrong-year state curve (refitted)	S1-wy vs S1	0.1185	0.0971	+22.1	[+20.1, +24.2]	[+15.1, +31.4]	0/12	clearly worse
Wrong-year signals at test only	S3-wyTest vs S3	0.1320	0.0963	+37.2	[+34.0, +40.5]	[+26.9, +49.9]	0/12	clearly worse

Scaled INLA: per-municipality rows are not stored, so only the year-level interval is given. Rows 1, 4 and 5 come from the official climate run (same folds and municipalities).

Table 6: Google Trends gain by region (design A, paired municipality bootstrap).

Region	Towns	B4s	S2 (Trends)	Change %	95% CI towns	Years better
Center-West	45	0.1276	0.1113	-12.8	[-14.3, -11.2]	11/12
North	45	0.1538	0.1509	-1.9	[-3.2, -0.7]	9/12
Northeast	45	0.1238	0.1171	-5.4	[-6.6, -4.4]	10/12
South	45	0.1223	0.0949	-22.4	[-25.8, -19.6]	11/12
Southeast	120	0.1022	0.0839	-17.9	[-19.4, -16.4]	12/12

Table 7: Gain by epidemic size. Strata are terciles of a municipality-year's annual total divided by that municipality's median total over the test years (a descriptive split, not used by any model).

Year size (tercile)	n	Trends %	CI towns	Weather %	CI towns
small year	1065	-8.4	[-9.5, -7.3]	+0.5	[-0.4, +1.5]
typical year	1064	-12.9	[-14.5, -11.2]	-0.4	[-1.5, +0.9]
large year	1065	-17.9	[-19.9, -15.9]	-4.4	[-6.0, -2.8]

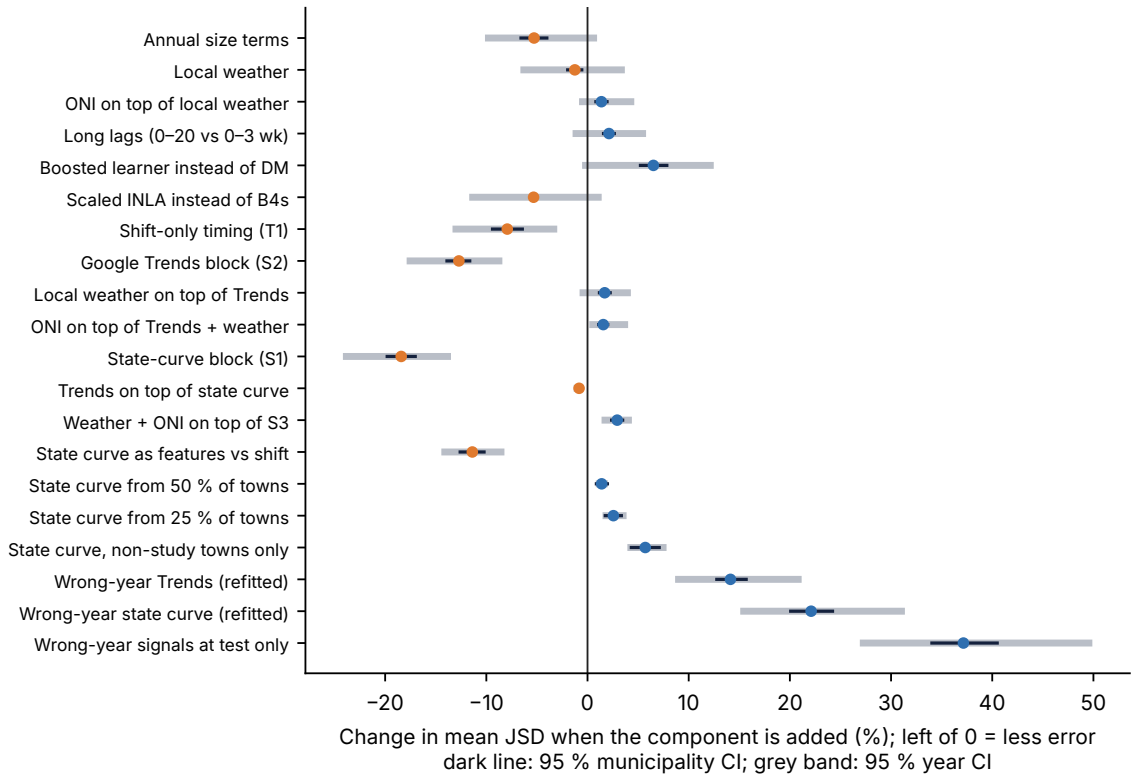


Figure 7: Ablation forest plot: relative change in mean JSD when each component is added (dot), with 95 % municipality (dark) and year (grey) bootstrap intervals.

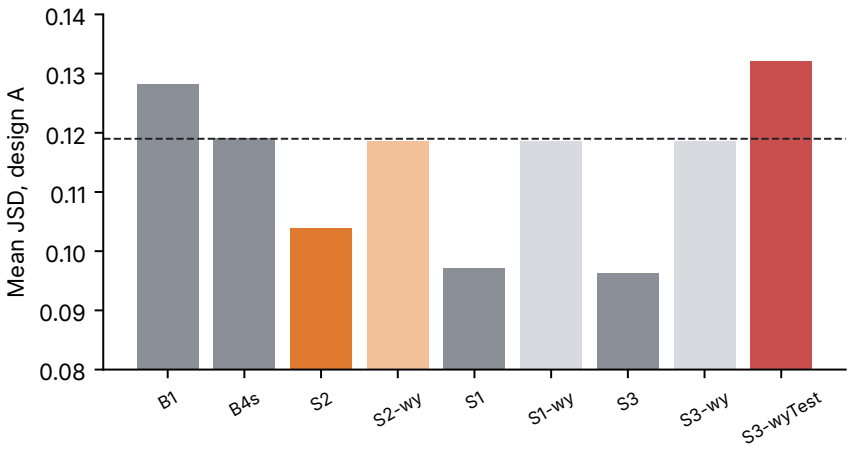


Figure 8: Wrong-year controls (light bars) against the correct-year models; dashed line: B4s.

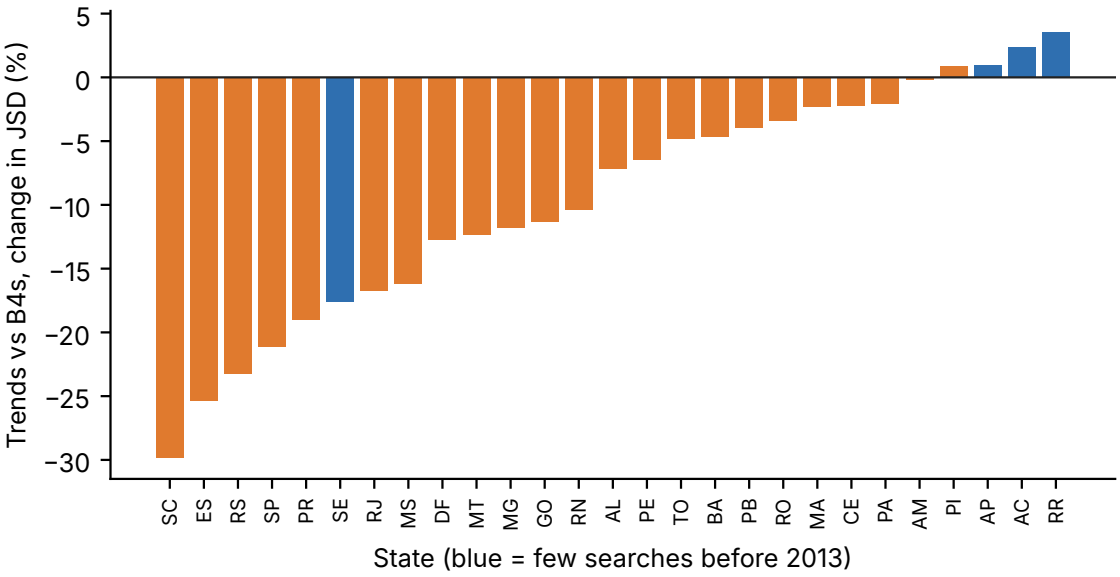


Figure 9: Trends vs B4s by state (design A), ordered by gain; blue: states with few searches before 2013.

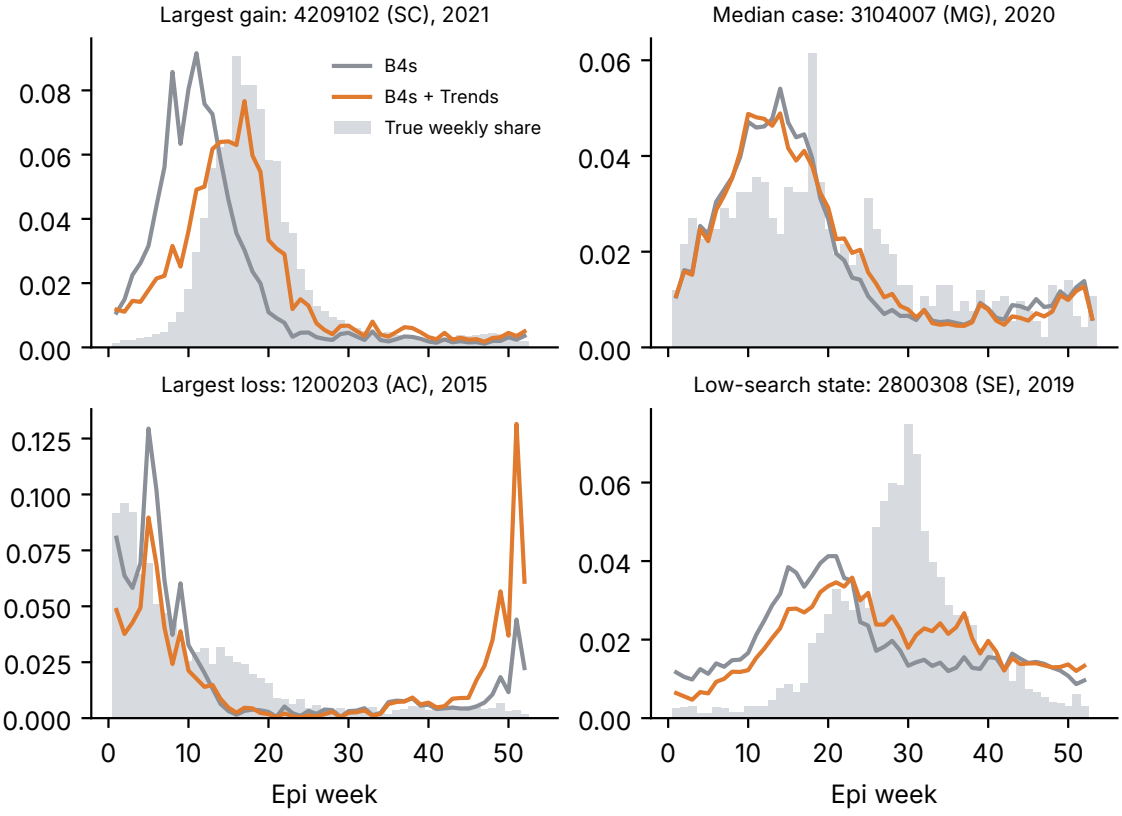


Figure 10: True weekly shares (grey) and rebuilt curves for four municipality-years selected by rule: largest gain, median case, largest loss, and largest gain in a low-search state.